Federal Funding Programs for Materials Informatics, Uncertainty Quantification, and Computational Materials Science Infrastructure: A Comprehensive Review (2025–2026)

1. Materials Genome Initiative: Strategic Framework and Current Status

1.1 MGI Vision and Federal Coordination

The **Materials Genome Initiative (MGI)** represents the United States' most ambitious federal investment in accelerating materials discovery and deployment, with cumulative federal investment exceeding **$250 million since its 2011 launch** and substantially larger indirect investments through aligned agency programs (The White House) . The initiative's foundational premise rests on exploiting transformational advances in computing capabilities, theoretical modeling, artificial intelligence and machine learning, and data mining to compress the typical **20-year timeline from materials discovery to market deployment** to approximately 10 years (National Academy of Engineering) .

The **White House Office of Science and Technology Policy (OSTP)** provides strategic leadership for MGI through the **National Science and Technology Council (NSTC) Subcommittee on the Materials Genome Initiative**, which coordinates activities across the National Science Foundation, Department of Energy, National Institute of Standards and Technology, Department of Defense, and National Aeronautics and Space Administration (mgi.gov) . This interagency coordination mechanism ensures complementary investments and prevents fragmentation of infrastructure development. The 2021 MGI Strategic Plan established three overarching goals that continue to guide federal investments: **unifying the Materials Innovation Infrastructure (MII), harnessing the power of materials data, and educating, training, and connecting the materials research and development workforce** (mgi.gov) .

The **2025–2026 strategic priorities** reflect a fundamental evolution toward **AI integration and autonomous experimentation**. Federal agencies are actively collaborating to "vastly accelerate the discovery to deployment timeline for new materials by integrating theory, data, experiment, and computation—including AI and autonomous experimentation" (The White House) . This represents a maturation from MGI's initial emphasis on data sharing and high-throughput computation to encompass **self-driving laboratory systems** that can independently design, execute, and interpret experiments with minimal human intervention (National Academy of Engineering) .

1.2 MGI Funding Landscape and Metrics

The **2025–2026 agency allocations** demonstrate sustained multi-agency commitment to MGI-aligned research:

Agency/Program	Annual Investment	Key Activities
NSF DMREF	**$50 million+**	Closed-loop materials discovery, autonomous experimentation (The White House)
DOE BES Computational Materials Sciences	**$30 million+**	Exascale-ready software, validated databases (HigherGov)
NIST Materials Measurement Laboratory	**~$15 million**	Standards, reference data, autonomous lab coordination (The White House)

Table 1 – continued

Agency/Program	Annual Investment	Key Activities
DARPA SURGE/PRIME	**$10.3 million** (2022–2026)	Additive manufacturing qualification with UQ (3Dnatives)

The **NSF DMREF program achieved a significant milestone in 2025**: over **4,000 publications and 100,000 citations**, corresponding to a knowledge-generation rate of approximately **one publication per day and one citation per hour** (The White House) . This productivity metric demonstrates the return on federal investment in coordinated materials informatics infrastructure.

International partnerships have expanded substantially, with the 2025 DMREF solicitation enabling collaboration with funding agencies in **Canada (NSERC), Germany (DFG), India (DST), and Israel (BSF)**, as well as the United States-Israel Binational Science Foundation (NSF - National Science Foundation) . These partnerships extend the reach of U.S. investments while promoting global standards for data sharing and interoperability. Complementary international initiatives include the **European Union's Horizon Europe Materials and Manufacturing program**, **Japan's NIMS Materials Data Platform**, and **Korea's MGI-K initiative** (mgi.gov) .

1.3 MGI Infrastructure and Data Ecosystems

The **Materials Innovation Platforms (MIPs)** program represents NSF's flagship investment in distributed research infrastructure, providing **$15–25 million over five years** to establish platforms integrating synthesis, characterization, and computation with comprehensive data science and cyberinfrastructure development (DMREF) . These platforms operate as national facilities with external user programs, enabling researchers nationwide to access cutting-edge capabilities without local infrastructure investment.

Platform	Institution	Focus Area	Status
2D Crystal Consortium	Pennsylvania State University	Two-dimensional materials	Operational (DMREF)
PARADIM	Cornell University / Johns Hopkins University	Oxide interfaces and heterostructures	Operational (DMREF)
Third MIP competition	TBD	Alloys, amorphous materials, composites	2025 awards (DMREF)

National data repositories form the backbone of MGI data infrastructure:

- **Materials Project** (DOE BES/NERSC): **150,000+ computed materials, 100,000+ experimental structures**, with pymatgen API and software ecosystem (mgi.gov)
- **AFLOW** (Duke University): High-throughput DFT with online discovery tools (mgi.gov)
- **OQMD** (Northwestern University): Open Quantum Materials Database with structure-property relationships (mgi.gov)
- **NIST Materials Data Repository**: Standard reference data and interoperability standards (mgi.gov)

FAIR data standards implementation remains a critical priority, with NIST leading development of **ontologies and interoperability frameworks** that enable cross-database search and analysis. The

2024 MGI Challenges specifically target unification of autonomous experimentation platforms and associated data infrastructure (mgi.gov) .

2. Department of Energy Programs

2.1 Advanced Scientific Computing Research (ASCR) Programs

2.1.1 Exascale Computing Project and Materials Science Applications The **Exascale Computing Project (ECP)**, which concluded its formal development phase in 2023, established the software ecosystem and application codes that enable materials simulations at unprecedented scale. The **ASCR Leadership Computing Challenge (ALCC)** program provides competitive access to DOE's most powerful supercomputing resources for materials science applications. A representative **2025–2026 ALCC award** titled **"Integrated Exascale Computational Workflows for Accelerated Material Synthesis"** exemplifies the program's relevance to the full materials development pipeline (U.S. DOE Office of Science) .

The **Leadership Computing Facility partnerships** provide essential infrastructure:

Facility	Location	System	Capabilities
ALCF	Argonne National Laboratory	Aurora	Exascale, AI-optimized (Department of Energy)
OLCF	Oak Ridge National Laboratory	Frontier	Exascale CPU/GPU (Department of Energy)
NERSC	Lawrence Berkeley National Laboratory	Perlmutter	Data-intensive science (mgi.gov)

2.1.2 ASCR Uncertainty Quantification Methodologies for Extreme-Scale Science ASCR has maintained **sustained investment in uncertainty quantification** spanning more than a decade. The **historical FOA DE-FOA-0000895 (2013)** established foundational support for UQ mathematics, with the program noting that "advances in these methods have contributed as much, if not more, to gains in computational science than hardware improvements alone" (Department of Energy) .

The **FY 2026 ASCR budget request** explicitly continues support for **multi-fidelity methods, Bayesian inference, and surrogate modeling** within its Applied Mathematics portfolio (Department of Energy) . Research thrusts include:

- **Scalable algorithms and libraries** for extreme-scale systems
- **Multiscale and multi-physics modeling** with quantified uncertainty
- **Methods that facilitate building and understanding foundational models** for AI capabilities
- **Efficient data analysis** at the edge of experiments and instruments (Department of Energy)

Awardee examples include **Sandia National Laboratories** and **Lawrence Livermore National Laboratory**, which have developed large-scale UQ software frameworks that serve broad scientific communities (SPIRAL Project) .

2.1.3 EXPRESS: 2025 Exploratory Research for Extreme-Scale Science The **EXPRESS program (DE-FOA-0003545)** provided **$16 million for approximately 20 awards** ranging from **$200,000 to $1,000,000 over two years**, with focus areas including **data management, visualization, and analytics for extreme-scale systems** (U.S. DOE Office of Science) . The relevance to **browser-based scientific visualization tools** emerges from the program's emphasis on innovative approaches to scientific computing challenges. The **January 2025 application deadline has passed**, with next anticipated cycle in **2026** (U.S. DOE Office of Science) .

2.2 Basic Energy Sciences (BES) Computational Materials Sciences

2.2.1 Computational Materials Sciences Program The **Computational Materials Sciences program (FOA DE-FOA-0001276)** supports **integrated theory-computation-experiment teams** developing **open-source community codes and validated databases for functional materials design** (HigherGov) . Key requirements include:

- **Exascale-ready software development** with deployment on leadership computing facilities
- **Validation and verification** against experimental data
- **Open-source release** with community support infrastructure

Award parameters: **\$2–3 million per year for 3–5 years**, with 2–4 awards per competition cycle (FederalGrants.com) . **Representative awardees** include principal investigators at **MIT, Caltech, University of Michigan, and national laboratories** (HigherGov) .

2.2.2 Computational Chemical Sciences Program The **Computational Chemical Sciences (CCS) program** provides complementary support for **open-source chemical simulation software**, with emphasis on **AI-ready datasets and machine learning interatomic potentials** (HigherGov) . The **FY 2026 request** anticipates ~**\$19 million** for 3-year projects, with award ceilings of **\$2 million per year** (HigherGov) . Integration with **Materials Project and NOMAD repositories** ensures community access to developed capabilities (mgi.gov) .

2.2.3 Energy Materials Network and Critical Materials Institute These **applied research programs** bridge fundamental computational science with technology transition. The **Critical Materials Institute** employs **high-throughput computational screening** to identify alternatives to supply-constrained elements, while the **Energy Materials Network** coordinates national laboratory capabilities with industry needs (mgi.gov) .

2.3 DOE Small Business Innovation Research (SBIR/STTR)

The **DOE SBIR/STTR programs** provide **Phase I awards of \$200,000–\$300,000** for proof-of-concept development and **Phase II awards up to \$1 million** for prototype development (ARPA-E) . **Topic areas** explicitly include **AI/ML for materials discovery, quantum materials, and advanced manufacturing**.

The **CATALCHEM-E SBIR/STTR program**, with full applications due **January 26, 2026**, targets **AI/ML-accelerated catalyst development** with the ambitious goal of completing **"10–15 years of traditional catalysis R&D work within 12–18 months"** (ARPA-E) . This exemplifies the transformative potential of materials informatics when effectively implemented. The **Technology Commercialization Fund** provides follow-on pathways for transitioning SBIR/STTR-developed technologies to market (ARPA-E) .

3. National Science Foundation Programs

3.1 Designing Materials to Revolutionize and Engineer our Future (DMREF)

3.1.1 DMREF as NSF's Primary MGI Mechanism The **DMREF program** represents **NSF's flagship MGI investment**, with **2025 funding exceeding $50 million** for new awards supporting researchers across **25 states** (The White House) . The **2025 solicitation (NSF 25-521)** maintains the program's established structure:

Parameter	Value
Award size	**$1.5–$2.0 million**
Duration	**4 years**
Team structure	Minimum 2 Senior/Key Personnel with complementary expertise (NSF - National Science Foundation)
Partner agencies	DOE-EERE, ONR, AFRL, NIST, Army DEVCOM (NSF - National Science Foundation)

The **closed-loop integration of theory, computation, and experiment** remains central, with **2025–2026 priorities** explicitly emphasizing **autonomous materials discovery and self-driving laboratories** (DMREF) .

3.1.2 DMREF Research Themes and Methodologies **Uncertainty quantification and error propagation in multi-scale models** has emerged as a critical capability area. DMREF projects address **quantification of uncertainty in machine learning models**, **propagation across atomistic-to-continuum scales**, and **integration with experimental validation** (DMREF) . The **data infrastructure** requirements encompass **open databases, APIs, and analysis tools** with explicit sustainability planning (FederalGrants.com) .

3.1.3 DMREF-Funded Infrastructure and Tools

Tool/Platform	Function	DMREF Contribution
pymatgen	Materials analysis library	Core development and maintenance (mgi.gov)
ASE	Atomistic simulations	Workflow integration (mgi.gov)
OVITO	Visualization and analysis	Feature development (mgi.gov)
Zentropy	Thermodynamic analysis	Specialized capabilities (mgi.gov)
Materials Project web interface	Browser-based discovery	Underlying methodology (mgi.gov)
AFLOW-online	Interactive materials search	Methodological advances (mgi.gov)

Multi-fidelity frameworks including **active learning and Bayesian optimization workflows** enable efficient exploration of vast materials design spaces (DMREF) .

3.1.4 Representative DMREF Projects and Institutions

Institution	Project Focus	Funding/Partners
University of Michigan	PRIME: AI/ML for metal additive manufacturing	**$10.3M, DARPA co-funding** (3Dnatives)
Brown University	Multi-fidelity UQ for materials design	DARPA EQUiPS foundation (Military Embedded Systems)
Stanford University	Supersonic nozzle materials with UQ	EQUiPS validation (SPIRAL Project)
Georgia Tech	Uncertainty quantification in phase-field models	Mesoscale UQ (DMREF)

3.2 Materials Innovation Platforms (MIP)

3.2.1 MIP Program Structure and Goals The **MIP program** provides **mid-scale infrastructure investments of $18–30 million over six years** to establish **distributed research platforms** with integrated synthesis, characterization, computation, and cyberinfrastructure (GrantExec) . The **2025 third competition** targeted **alloys, amorphous materials, and composites**, with **1–3 awards anticipated from $30 million total program funding** (GrantExec) .

3.2.2 Current and Anticipated MIP Competitions

Competition	Year	Focus Area	Status
First MIP	2015	2D materials, interfaces	Complete (DMREF)
Second MIP	2019	Biomaterials, polymers	Complete (FederalGrants.com)
Third MIP	**2025**	**Alloys, amorphous, composites**	**Awards pending** (GrantExec)
Fourth MIP	**Anticipated 2026–2027**	TBD	**Planning stage** (DMREF)

3.2.3 MIP Cyberinfrastructure and Open Science MIP platforms emphasize **real-time data streaming and remote access capabilities**, enabling worldwide researcher participation without physical presence (DMREF) . **Integration with national supercomputing centers** ensures computational modeling keeps pace with experimental data generation, while **training programs** develop workforce capabilities in data-intensive materials science (DMREF) .

3.3 Cyberinfrastructure for Sustained Scientific Innovation (CSSI)

The **CSSI program** supports **research software infrastructure** with awards ranging from **$500,000 to $2 million over 3–5 years** (DMREF) . For materials science, CSSI provides essential funding for:

- **Open-source materials informatics software** development and maintenance
- **Data lifecycle management and preservation tools**
- **Browser-based visualization and analysis platforms** (DMREF)

The **sustainability focus** distinguishes CSSI from project-specific software development, ensuring that valuable tools receive long-term support beyond initial research grants (DMREF) .

3.4 NSF SBIR/STTR for Materials Startups

Phase	Funding	Duration	Purpose
Phase I	**Up to $305,000**	6–18 months	Proof-of-concept (BW&CO)
Phase II	**Up to $1,250,000**	24 months	Prototype development (BW&CO)
Supplemental	**$500,000+**	Variable	Additional milestones (BW&CO)

Topic areas include **Advanced Materials (AM), Advanced Manufacturing (M), Artificial Intelligence (AI), and Cloud/High-Performance Computing (CH)** (BW&CO) . **Materials informatics-focused awardees** include companies developing **uncertainty quantification software, materials databases and knowledge graphs, and visualization/VR tools for materials science** (DMREF) .

4. Defense Advanced Research Projects Agency Programs

4.1 DARPA Defense Sciences Office (DSO) Materials Programs

4.1.1 Enabling Quantification of Uncertainty in Physical Systems (EQUiPS)

Attribute	Details
Duration	**2014–2019** (Military Embedded Systems)
Core objective	Make UQ routine in complex system design (Military Embedded Systems)
Key innovation	Multi-fidelity methods for order-of-magnitude cost reduction (Military Embedded Systems)

EQUiPS established foundational capabilities for **combining information from models of varying accuracy and computational cost**, with validation in demanding aerospace applications (Military Embedded Systems) .

4.1.2 EQUiPS Technical Approaches and Outcomes

Institution	Application	Outcome
Brown University	Supersonic nozzle materials	Multi-fidelity UQ framework validated (Military Embedded Systems)
Stanford University	Hydrofoil vessel design	Uncertainty bounds for extreme sea states (SPIRAL Project)
Sandia National Labs	General UQ software	Large-scale frameworks transitioned (SPIRAL Project)

The **Brown University** research specifically addressed **materials design under uncertainty** where "the number of parameters can be in the thousands and design requires accounting for uncertain operating conditions, novel materials whose behavior may not be fully understood, and manufacturing imperfections" (Military Embedded Systems) .

4.2 Structures Uniquely Resolved to Guarantee Endurance (SURGE)

4.2.1 SURGE Program Scope and Objectives The **SURGE program** addresses **additive manufacturing part qualification** through **real-time process monitoring and predictive modeling**, targeting the fundamental challenge that layer-by-layer fabrication creates complex, process-dependent microstructures that resist traditional quality assurance approaches (3Dnatives) .

4.2.2 Predictive Real-Time Intelligence for Metal Endurance (PRIME)

Attribute	Details
Lead institution	**University of Michigan** (3Dnatives)
Funding	**$10.3 million** (3Dnatives)
Duration	**2022–2026** (3Dnatives)
PI	**Veera Sundararaghavan** (Engineering Research News)

Technical architecture:

Component	Institution/Partner	Function
Multi-sensor data collection	University of Michigan	Capture thermal history, melt pool dynamics (3Dnatives)
Digital twin modeling	**AlphaSTAR** (industry)	Physics-based process simulation (Metal AM magazine)
Microstructure modeling	University of Michigan	Crystal grains, phases, pores (Engineering Research News)
Uncertainty quantification	**UC San Diego**	**Predict part resilience with confidence bounds** (Engineering Research News)
Experimental validation	**Auburn University**	Fatigue testing to failure (Engineering Research News)

The **PRIME vision** extends to **platform-agnostic prediction**: "If PRIME takes off, it's like giving 3D printing a crystal ball—predicting the lifetime of LPBF parts across platforms and turning critical part production into a low-cost, distributed dream" (Engineering Research News) . **Industrial transition** is facilitated by **ASTM International** partnership and industry collaborators **Addiguru** and **AlphaSTAR** (3Dnatives) .

4.3 Other DARPA Materials and Manufacturing Programs

Program	Focus	Relevance to Materials Informatics
Open Manufacturing Program	Process qualification	Computational prediction for certification

Table 13 – continued

Program	Focus	Relevance to Materials Informatics
Materials Development for Platforms (MDP)	Rapid materials optimization	AI-driven design workflows
DSO Office-wide BAA	Broad defense science	Rolling opportunities for innovative proposals (yale.edu)

5. Advanced Research Projects Agency-Energy Programs

5.1 ARPA-E Advanced Materials Portfolio

5.1.1 MAGNITO: Magnetic Materials for Energy Applications

Attribute	Details
Focus	AI/ML-accelerated discovery of **rare-earth-free permanent magnets** (ARPA-E)
Approach	High-throughput computation + combinatorial synthesis (grantedai.com)
Awardees	National labs, universities, industry teams (grantedai.com)

MAGNITO addresses critical supply chain vulnerabilities while demonstrating practical value of computational materials discovery workflows (grantedai.com) .

5.1.2 CATALCHEM-E: Catalyst Development with AI

Attribute	Details
Launch	**January 2026** (ARPA-E)
Mechanism	SBIR/STTR (ARPA-E)
Goal	**10–15 years of traditional R&D in 12–18 months** (ARPA-E)
Methods	AI/ML + high-throughput experimentation (ARPA-E)

5.1.3 Historical Programs with Informatics Components

Program	Year	Focus	Legacy
ENLITENED	2017	Thermal management materials	Computational screening methods (Department of Energy)
PNDIODES	2017	Power electronics materials	High-throughput optimization (Department of Energy)
OPEN+/FOCUS	Various	Foundational energy research	Flexible funding for emerging methods

5.2 ARPA-E Technology-to-Market Pathways

ARPA-E's explicit **Technology-to-Market** mandate distinguishes its approach from foundational research agencies. **SBIR/STTR commercialization support**, **industry partnership facilitation**, and **demonstration project coordination** ensure that materials informatics investments have clear pathways to real-world impact (ARPA-E) .

6. National AI Research Resource and AI-Materials Workflows

6.1 NAIRR Pilot Program (2024–2026)

Attribute	Details
Authorization	**$2.6 billion under CHIPS and Science Act** (The White House)
Lead agency	**National Science Foundation** (The White House)
Status	**Pilot phase (2024–2026)** (The White House)

Resources for materials science (The White House) :

Resource Type	Specific Capabilities
Computational	GPU clusters, cloud credits for large model training
Datasets	Curated materials science corpora
Software/Tools	AI/ML platforms, workflow orchestration systems

Access mechanisms include pilot user portal with proposal-based allocation, training and educational resources, and integration with existing materials databases (The White House) . However, **specific allocation details and materials science user statistics remain limited in publicly available documentation**.

6.2 AI-Materials Workflow Development

Federal investments are converging on **autonomous experimentation pipelines** that integrate:

- **AI-driven experiment design** (Bayesian optimization, active learning)
- **Robotic synthesis and characterization** (self-driving laboratories)
- **Real-time data analysis and model refinement** (closed-loop learning)

Foundation models for materials—large neural networks trained on diverse materials data—represent an emerging frontier with potential to transform prediction capabilities. **Multi-modal data fusion** combining text, structure, spectra, and properties enables richer representations than any single data type alone (The White House) .

7. Multi-Fidelity Uncertainty Quantification Frameworks

7.1 Federal Investment in Multi-Fidelity UQ

Agency/Program	Contribution	Timeframe
DOE ASCR Applied Mathematics	Mathematical foundations, extreme-scale algorithms	2013–present (Department of Energy)
DARPA EQUiPS	Engineering validation, cost reduction demonstrations	2014–2019 (Military Embedded Systems)
NSF DMREF	Materials-specific implementations, experimental integration	2012–present (DMREF)
DARPA SURGE/PRIME	Real-time qualification with quantified uncertainty	2022–2026 (3Dnatives)

This **distributed investment pattern**—foundational mathematics through ASCR, engineering validation through DARPA, materials-specific implementation through NSF—represents a **coordinated federal strategy** that avoids duplication while ensuring comprehensive capability development.

7.2 Technical Approaches and Software

Method	Description	Open-Source Implementation
Gaussian process regression	Probabilistic surrogate modeling	GPyTorch (Military Embedded Systems)
Bayesian multi-fidelity modeling	Information fusion across fidelity levels	MUQ, Emukit (Military Embedded Systems)
Active learning/adaptive sampling	Optimal experimental design	PyMC, custom implementations (Military Embedded Systems)

7.3 Application Domains

Scale	Methods	Example Applications
Atomistic	DFT UQ, force field calibration	Electronic structure, defect energetics (Military Embedded Systems)
Mesoscale	Phase-field UQ, crystal plasticity	Microstructure evolution, mechanical response (DMREF)
Macroscale	Structural UQ, digital twins	Component qualification, lifespan prediction (Engineering Research News)

8. Open-Source Materials Databases and Infrastructure

8.1 Federally Supported Database Initiatives

Database	Host/Lead	Scale	Federal Support
Materials Project	LBNL/NERSC	**150,000+ computed, 100,000+ experimental** (mgi.gov)	DOE BES core + NERSC allocation
AFLOW	Duke University	High-throughput DFT	NSF, DOE (mgi.gov)
OQMD	Northwestern University	Complementary DFT coverage	NSF (mgi.gov)
NIST Materials Data Repository	NIST	Standard reference data	NIST appropriations (mgi.gov)

2025–2026 development priorities for Materials Project include **autonomous workflows and machine learning interatomic potentials** that extend predictive capabilities to kinetic and finite-temperature properties (mgi.gov) .

8.2 Database Sustainability and Governance

Challenge	Response Strategy
Core operational funding	Facility status (NERSC), agency core support (NIST)
Project-specific enhancements	Research grants (DMREF, CMS)
Community contribution	Open-source governance (pymatgen, ASE)
Long-term sustainability	Foundation partnerships, industry consortia (mgi.gov)

9. Browser-Based Scientific Visualization Tools

9.1 Federal Support for Visualization Infrastructure

Program/Agency	Mechanism	Focus
NSF CSSI	Software infrastructure awards	Web-native visualization development (DMREF)
DOE ASCR	Data and visualization programs	Extreme-scale rendering (U.S. DOE Office of Science)
NIST	Measurement science	Materials characterization visualization (mgi.gov)

9.2 Representative Platforms and Tools

Platform	Capabilities	Access
Materials Project web interface	Structure visualization, band structures, phase diagrams	materialsproject.org (mgi.gov)
AFLOW-online	Interactive materials discovery, property search	aflow.org (mgi.gov)

Table 25 – continued

Platform	Capabilities	Access
OVITO Web	Browser-based trajectory analysis for MD simulations	ovito.org (mgi.gov)
NGLView / 3Dmol.js	Embeddable molecular structure rendering	Jupyter integration (mgi.gov)

9.3 Emerging Directions

- **Virtual and augmented reality** for immersive materials exploration
- **Real-time collaborative visualization** for distributed research teams
- **AI-generated insights integration** for automated annotation and guidance

10. Complementary International and Private Initiatives

10.1 International Programs

Country/Region	Program	Key Features	Relevance to U.S. MGI
United Kingdom	**National Materials Innovation Strategy** (2025)	**£2M initial funding**, "Materials 4.0" infrastructure, Innovate UK grants (biforesight.com)	Transatlantic collaboration opportunity
European Union	Horizon Europe Materials and Manufacturing	Open data, FAIR principles, Materials Genome Europe coordination (mgi.gov)	Standards harmonization
Japan	NIMS Materials Data Platform	Comprehensive experimental database (mgi.gov)	Data sharing agreements
Korea	MGI-K, KIST materials informatics	National AI-materials programs (mgi.gov)	Competitive benchmarking

The **UK National Materials Innovation Strategy** explicitly diagnoses uneven distribution of "Materials 4.0" capabilities and proposes shared national infrastructure—paralleling challenges and responses in the U.S. context (biforesight.com) .

10.2 Private Sector and Philanthropic Initiatives

Organization	Contribution	Materials Science Relevance
Chan Zuckerberg Initiative	Open-source scientific software (Matplotlib, Jupyter) (mgi.gov)	Foundational visualization and analysis tools
Schmidt Futures	AI for science programs (mgi.gov)	Methodological advances, talent development

Table 27 – continued

Organization	Contribution	Materials Science Relevance
Industry consortia (MRS, ASM International)	Professional community, standards development	Technology transition pathways

10.3 Public-Private Partnerships

Partnership Type	Examples	Function
Industry alliances	SEMI Smart Manufacturing, AIP alliances	Pre-competitive research coordination (mgi.gov)
National lab entrepreneurship	Argonne, LBNL, ORNL embedded programs	Technology commercialization (mgi.gov)
Federal-industry cost-sharing	ARPA-E, SBIR/STTR matching	Risk-sharing for early-stage development

11. Program Status Summary and Future Outlook

11.1 Currently Open and Anticipated Programs (March 2026)

Program	Agency	Status	Anticipated Timeline
NSF DMREF	NSF	Annual solicitation	**2026 competition anticipated** (DMREF)
NSF MIP fourth competition	NSF	Planning stage	**2026–2027 announcement** (DMREF)
DOE BES Computational Materials Sciences	DOE	Cyclical FOA	**Next competition TBD** (HigherGov)
ARPA-E OPEN+	ARPA-E	Rolling topics	**Ongoing** (grantedai.com)
ARPA-E CATALCHEM-E	ARPA-E	**Full applications due January 26, 2026**	**Active** (ARPA-E)

11.2 Recently Closed Programs (2024–2025)

Program	Closure	Outcomes/Transition
DOE ASCR EXPRESS 2025	January 2025 deadline (U.S. DOE Office of Science)	Awards made; next cycle anticipated 2026
NSF CSSI February 2025	February 2025 (DMREF)	Review complete; awards pending
NSF MIP third competition	May 2025 deadline (GrantExec)	Awards pending announcement
DARPA SURGE/PRIME	**Nearing completion 2026** (3Dnatives)	**Results transition to defense/ commercial applications**

11.3 Anticipated Future Directions

Direction	Drivers	Key Programs
AI-augmented autonomous materials discovery	MGI workshops, NAIRR pilot, self-driving lab advances (mgi.gov)	DMREF, MIP, NAIRR
Quantum computing for materials simulation	Hardware advances, algorithm development	ASCR, BES exploratory programs
Digital twins and real-time qualification	Industrial demand, PRIME validation (Engineering Research News)	DARPA transition, SBIR/STTR
International coordination and data sharing	Supply chain security, global challenges (biforesight.com)	MGI partnerships, bilateral agreements

The **federal funding landscape for materials informatics, uncertainty quantification, and computational materials science infrastructure** demonstrates **substantial sustained investment with coherent multi-agency coordination** through the MGI framework. While specific program details evolve annually, the foundational commitment to **accelerating materials discovery through integrated computational-experimental approaches with rigorous uncertainty quantification** appears firmly established for 2025–2026 and beyond. The convergence of **AI/ML capabilities, autonomous experimentation, and extreme-scale computing** creates unprecedented opportunities for transformative advances, with federal investments positioned to capture these opportunities through strategic coordination across foundational research, applied development, and commercialization pathways.